Mark Julius P. Mella

Software Engineer | Agentic AI Engineer
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Professional Summary

Software engineer with 6+ years across freelance and product development, building responsive applications with Vue, React, Next.js, TypeScript, and JavaScript. Experienced in reusable components, state management, REST APIs, real-time integrations, and connecting interfaces with backend services and persistent data. Improve performance and SEO through caching, code splitting, lazy loading, and image and font delivery. Use Git, code review, and CI/CD for maintainable delivery while transitioning into agentic AI engineering through scoped AI-assisted workflows with human review and validation.

Work Experience

Software Developer — Frontend & API Integration
May 2023 - Present
DigiPlus Interactive Corp

- Integrate sports data, account and customer-support REST APIs and real-time services for ArenaPlus, connecting web and mobile interfaces to backend data.
- Build deposit, withdrawal, linked-wallet and transaction-history interfaces with loading, pending and error states so users can follow payment progress.
- Scaffold Vue 3 and TypeScript applications with authenticated routing, shared layouts, stores and services, providing a reusable foundation for new features.
- Keep noncritical SDKs and below-the-fold content off the initial render through deferred loading, code splitting and image/font delivery changes.
- Refine persistent caching, stale-request handling and session cleanup to prevent outdated responses and retained resources from carrying into later journeys.
- Contribute to Vue 2-to-Vue 3 migrations and shared responsive components, keeping layouts and theme behavior consistent across product features.

Frontend Developer
May 2022 - May 2023
Solid Leisure Solutions

- Delivered UI updates and fixes for the BingoPlus digital bingo platform.
- Built reusable components from Figma designs, adapting layouts to desktop and mobile.
- Integrated REST APIs to connect frontend features with backend data and service responses.
- Embedded customer-support chat so users could access assistance within the application.
- Refined frontend performance to keep application interactions responsive.

Technical Skills

Frontend engineering: JavaScript, TypeScript, Vue 2 / Vue 3, React, Next.js, HTML, CSS, Tailwind CSS

API integration and state: REST APIs, Axios, Pinia, Vuex, TanStack Vue Query, SignalR, Mobile WebViews

Performance and SEO: Lighthouse, Caching, IndexedDB, Lazy loading, Code splitting, Search engine optimization (SEO)

AI-assisted engineering: Codex, ChatGPT, Cursor, Claude Code, AI-assisted research, Task decomposition, Output validation

Version control and delivery: Git / GitHub, Pull requests, CI/CD (continuous integration and delivery), Vite, Vercel

Projects

Frontend Architecture and Payment API Integration

Built authenticated routing, layouts, stores and services. Integrated four wallet journeys: deposits, withdrawals, linked accounts and transaction history, with request-state feedback and embedded navigation.

Vue 3, TypeScript, REST APIs, Pinia, Vite

Product Storefront and Inventory Management

Preserved loaded storefront data when refreshes fail through version-aware caching. Added subcategory controls, Firestore access rules and inventory cache refresh so updates reach the storefront.

React, Next.js, TypeScript, Firebase, IndexedDB

Catalog Editing Workspace

Saved catalog settings and item order in Firestore so edits survive reloads. Added optimistic updates with rollback on failed writes to keep the interface aligned with saved data.

React, Next.js, TypeScript, Firestore

Campaign Forms, Event Draws and Reporting

Refined receipt-upload, participation and draw-loading states. Added duplicate-receipt review, date filters and reporting so operators can inspect campaign submissions.

React, Vue 3, TypeScript, Firebase

Agentic AI Engineering Practice

AI study began in March 2025. Apply coding agents to scoped research, debugging and refactoring, with explicit constraints and human review.

- Use Codex and ChatGPT for research, and Cursor and Claude Code to audit code, debug defects and refactor in small, reviewable changes.
- Define scope, context, constraints and acceptance criteria so agent output can be checked against a specific result.
- Review Git changes and affected call sites; validate types, builds and browser behavior, and check CI/CD results before release.

Additional Work Experience

Frontend Developer - Internship

March - April 2022

Public Safety Savings and Loan Association, Inc. (PSSLAI)

- Built employee risk-calculation interfaces using React, Node.js and AWS, giving staff a web interface for risk assessment.

Freelance Frontend Developer

2020 - 2022

- Delivered responsive web and mobile layouts from Figma designs, with reusable components for consistent page structure.

Education

Bachelor of Science in Computer Science

2018 - 2022

Pamantasan ng Lungsod ng Pasig

Awards and Certifications

- Best in Thesis - AWIT, an interpreted Turing-complete scripting language written in C with bytecode generation and garbage collection.
- Dean's Lister, 2021-2022.